

Antiderivatives of Monomially Weighted Fabius-Type Functions

Exact Volterra Ladders, Newton–Mellin Series, Rvachev Folds, Fractional Orders, and Inverse-Quantile Integrals

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Abstract

Let F be the bounded Fabius distribution function, let up be Rvachev’s compactly supported up-function, and let $Q = F^{-1}$ on $[0, 1]$. This report constructs normalized antiderivatives of $x^p F(x)$ and their higher-order Volterra analogues. For $p \in \mathbb{N}_0$, the answer is a finite closed expression in dyadically contracted Fabius values. More generally, for every complex exponent α , every integer order $n \geq 1$, and $0 < x \leq 1$, we prove a locally uniformly and absolutely convergent Newton–Volterra expansion

$$\frac{1}{(n-1)!} \int_0^x (x-t)^{n-1} t^\alpha F(t) dt = \sum_{r \geq 0} (-1)^r \binom{n+r-1}{r} \alpha^r x^{\alpha-r} 2^{\binom{n+r}{2}} F\left(\frac{x}{2^{n+r}}\right).$$

Endpoint flatness makes this an entire function of α , including all negative powers; nonnegative integral exponents are exactly the terminating cases. At $x = 1$, the coefficients become the Fabius moments and give a Newton interpolation formula, a two-parameter family of exact moment identities, and a logarithmic constant

$$\int_0^1 \frac{F(x)}{x} dx = -\mathbb{E} \log X = \sum_{n \geq 1} \frac{d_n}{n} = \log 2 + \frac{1}{2} \sum_{n \geq 1} \frac{c_n}{n}.$$

We derive global piecewise formulas, exponential generating functions, Bell–Bernoulli cumulant forms, formulas for $F(1-x) = 1 - F(x)$, complete piecewise antiderivatives of $x^p \text{up}(x)$, exact higher derivatives in Thue–Morse form, and Riemann–Liouville extensions of nonintegral order. For the inverse Fabius function we obtain closed change-of-variable formulas, including

$$\int_0^y Q(u) du = yQ(y) - F\left(\frac{Q(y)}{2}\right), \quad \mathcal{I}^\nu Q(y) = \frac{1}{\Gamma(\nu+1)} \int_0^{Q(y)} (y - F(t))^\nu dt.$$

Combining the inverse endpoint equivalent already developed in the repository with regular-variation arguments yields new asymptotics for weighted inverse primitives and for large quantile moments. The final sections separate proved statements from conjectural saddle refinements, record reproducible numerical checks, and propose further formalization and arithmetic directions. The integer core is now machine checked in Lean: this comprises the normalized Volterra operator, its identification with literal repeated integration, the finite polynomial and natural-monomial commutators, the global signed Fabius ladder, and the bounded formulas on the full range $x \leq 1$. The infinite complex-power series, Mellin consequences, fractional orders, global exterior pieces, Rvachev folds, and inverse-function formulas remain research-frontier material.

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1 Scope, status, and corpus audit

1.1 The source corpus

The requested source tree was audited recursively at the repository snapshot identified in its own frontier inventory as commit 32d6d36c51d803289e6d6a0dc0c37753766eba47 (27 August 2026). That inventory records 67 distinct `*.tex` paths under `Analysis/FabiusFunction/docs`. The audit included the live primary exposition, the mathematical glossary, the Lean walkthrough, the non-elementarity article, the consolidated non-formalized frontier volume, the source-paper transcriptions, and the frozen archive and draft families [1, 2, 3, 4, 5, 6].

Archived files contain many historical copies of material later consolidated into the live exposition or frontier volume. For nonduplication, the present work therefore used two passes:

1. a path-complete inventory and phrase audit over the recursive corpus;
2. theorem-level comparison against the live primary exposition and the consolidated frontier volume, with targeted checks of the inverse, repeated-integration, Newton/Richardson, moment, asymptotic, and non-elementarity source families.

At the audited snapshot, the corpus already contained the basic functional equations, exact dyadic values, moment recurrences, the sinc product, all-orders endpoint asymptotics, repeated-averaging limits, and extensive inverse-function work. It did *not* yet develop the normalized antiderivative ladder of this report, the finite monomial formula, the arbitrary-complex-power Newton–Volterra series, or the inverse-primitive reductions below. Literal searches for “antiderivative” and “indefinite integral” returned no such treatment in the live primary or consolidated frontier source. This establishes novelty *relative to the audited repository*; it is not a claim that no equivalent identity has ever appeared elsewhere. In particular, the base functional integral identity $\int_0^x F(t) dt = F(x/2)$ is an immediate classical consequence of the defining equation. The subsequent Lean implementation described below changes the current formalization status, not this historical snapshot audit.

Status convention used in this report

Every item labeled theorem, proposition, lemma, or corollary is either proved in place, derived immediately from a proved identity, or explicitly identified as an imported standard theorem; that label alone does not assert a Lean proof. In particular, the first formula in [theorem 13.4](#) is the classical Lagrange inverse-derivative formula; the report cites that result and proves only the coefficient conversion used here. “Repository input” means that an already established result is imported from the audited corpus. “Conjecture” means that the statement is supported by the preceding analysis but is not proved. Besides repository inputs—including derivative identities already present in the audited corpus and a post-snapshot Lean proof of the inverse endpoint equivalent—only the declarations in the crosswalk below are claimed to be machine checked by the present status update; all other new claims in this report remain non-formalized.

1.2 Lean formalization update

The reusable foundation is formalized in `FabiusFunction.NormalizedVolterra`. Its definition `normalizedVolterra` takes values in an arbitrary real normed vector space and uses oriented interval integrals. For continuous Banach-valued inputs, `iteratedPrimitive_eq_normalizedVolterra` identifies its Cauchy kernel with literal repeated application of the first primitive. Thus the formal result is not confined to the scalar Fabius case or to ordered endpoints.

Mathematical content	Exact Lean declaration and module
Normalized kernel and literal-iteration bridge	<code>normalizedVolterra</code> , <code>iteratedPrimitive</code> , and <code>iteratedPrimitive_eq_normalizedVolterra</code> in <code>FabiusFunction.NormalizedVolterra</code> .
Kernel raising and arbitrary finite polynomial weights	<code>normalizedVolterra_kernel_pow</code> , <code>normalizedVolterra_polynomial</code> , and <code>normalizedVolterra_monomial</code> in <code>FabiusFunction.NormalizedVolterra</code> .
Signed Fabius primitive ladder, all orders and all real endpoints	<code>normalizedVolterra_extendedFabius</code> in <code>FabiusFunction.FabiusAntiderivatives</code> .
Bounded Fabius primitive ladder on the proved initial half-line $x \leq 1$	<code>normalizedVolterra_fabiusReal_of_le_one</code> in <code>FabiusFunction.FabiusAntiderivatives</code> .
Finite natural-monomial Fabius formulas	<code>normalizedVolterra_pow_mul_extendedFabius</code> and <code>normalizedVolterra_pow_mul_fabiusReal_of_le_one</code> in <code>FabiusFunction.FabiusAntiderivatives</code> .

The generic polynomial theorem permits every order $n \in \mathbb{N}_0$, every real base point and endpoint, every interval-integrable input with values in an arbitrary real normed vector space, and an arbitrary real polynomial. Its coefficients are Hasse derivatives and rising factorials, so the order-zero identity and all endpoint degeneracies require no side conditions. The Fabius specializations likewise include order zero. For the bounded function the proved dyadic formula holds on $(-\infty, 1]$, not merely on $[0, 1]$.

No claim of this update extends to the infinite complex-power Newton–Volterra series or its normal convergence, the Mellin/Newton and negative-power consequences, Riemann–Liouville fractional order, the $x > 1$ branches of the bounded global piecewise formula, folded Rvachev primitives or their Fourier renormalization, or any inverse-Fabius primitive or inverse asymptotic. Those remain explicit formalization frontiers.

1.3 What “closed form” means here

The repository proves strong non-elementarity statements for F , up , and Q [5, 2]. Consequently an antiderivative of $x^p F(x)$ cannot generally be elementary on any interior interval: if it were, its derivative divided by x^p on a subinterval not containing zero would make F elementary there. The meaningful native closed class is therefore the algebra generated by polynomials, rational constants, and finitely many dyadically scaled Fabius values. The finite formula in [theorem 5.1](#) is closed in precisely that sense.

2 Normalization and probabilistic infrastructure

2.1 The three Fabius-type functions

We follow the repository’s careful distinction between three objects. The bounded distribution function is

$$F(x) = 0 \quad (x \leq 0), \quad F(x) = 1 \quad (x \geq 1),$$

with

$$F(1 - x) = 1 - F(x), \quad x \in \mathbb{R}, \quad (1)$$

$$F'(x) = 2F(2x), \quad 0 \leq x \leq \frac{1}{2}. \quad (2)$$

The compactly supported Rvachev function is

$$\text{up}(x) = F(1 - |x|), \quad (3)$$

and the signed global extension is

$$\mathcal{F}(x) = \sum_{m \geq 0} (-1)^{s_2(m)} \text{up}(x - 2m - 1), \quad \mathcal{F}'(x) = 2\mathcal{F}(2x). \quad (4)$$

Here $s_2(m)$ is the binary digit sum. The bounded and signed functions coincide on $(-\infty, 1]$, but not globally [1, 11].

Let $(U_j)_{j \geq 1}$ be independent uniform random variables on $[0, 1]$, and put

$$X = \sum_{j \geq 1} 2^{-j} U_j. \quad (5)$$

Then F is the distribution function of X , and $X \stackrel{d}{=} 1 - X$. The density of $Y = 2X - 1$ is up . We use the moments

$$d_n = \mathbb{E}X^n, \quad c_n = \mathbb{E}(2X - 1)^{2n} = \int_{-1}^1 x^{2n} \text{up}(x) dx. \quad (6)$$

The repository establishes

$$d_0 = 1, \quad (n+1)(2^n - 1)d_n = \sum_{k=0}^{n-1} \binom{n+1}{k} d_k, \quad (7)$$

$$d_n = n! 2^{\binom{n}{2}} F(2^{-n}), \quad (8)$$

$$c_n = \frac{2d_{2n+1}}{2n+1}. \quad (9)$$

The moment generating function

$$G(z) = \mathbb{E}e^{zX} = \sum_{n \geq 0} d_n \frac{z^n}{n!} = \prod_{j \geq 1} \frac{e^{z/2^j} - 1}{z/2^j} \quad (10)$$

satisfies

$$G(2z) = \frac{e^z - 1}{z} G(z), \quad G(-z) = e^{-z} G(z). \quad (11)$$

At imaginary arguments, the centered transform $e^{-z} G(2z)$ becomes the familiar infinite sinc product for up [10, 9, 1].

2.2 Normalized integer and fractional primitives

For an integer $n \geq 1$, a real base point a , and an oriented endpoint x , define the normalized Volterra primitive

$$(\mathcal{J}_{n,a}f)(x) = \frac{1}{(n-1)!} \int_a^x (x-t)^{n-1} f(t) dt, \quad \mathcal{J}_{0,a}f = f. \quad (12)$$

The zero-based notation is $\mathcal{J}_n = \mathcal{J}_{n,0}$. For positive n , it is the unique solution of $D^n u = f$ whose derivatives of orders $0, \dots, n-1$ vanish at a . Every other n -fold antiderivative differs from it by a polynomial of degree at most $n-1$. For continuous Banach-valued inputs, the formal theorem `iteratedPrimitive_eq_normalizedVolterra` proves that (12) is exactly literal repeated integration.

For $\text{Re } \nu > 0$, the Riemann–Liouville version is

$$(\mathcal{I}^\nu f)(x) = \frac{1}{\Gamma(\nu)} \int_0^x (x-t)^{\nu-1} f(t) dt. \quad (13)$$

The notation $\mathcal{J}_n = \mathcal{I}^n$ will be used when the order is positive and integral. For a complex exponent α and positive x , we take $x^\alpha = \exp(\alpha \log x)$ with the real logarithm.

3 The dyadic antiderivative ladder

Theorem 3.1 (Exact normalized primitive ladder). *For every $n \in \mathbb{N}_0$ and every real $x \leq 1$,*

$$(\mathcal{J}_n F)(x) = 2^{\binom{n}{2}} F\left(\frac{x}{2^n}\right). \quad (14)$$

For the signed global extension, the same formula holds for every real x :

$$(\mathcal{J}_n \mathcal{F})(x) = 2^{\binom{n}{2}} \mathcal{F}\left(\frac{x}{2^n}\right). \quad (15)$$

The order-zero cases use the identity convention in (12).

Proof. The case $n = 0$ is immediate. For the signed extension set $H_n(x) = 2^{\binom{n}{2}} \mathcal{F}(x/2^n)$. The global dilation equation gives

$$H'_n(x) = 2^{\binom{n}{2}-n} \mathcal{F}'(x/2^n) = 2^{\binom{n}{2}-n+1} \mathcal{F}(x/2^{n-1}) = H_{n-1}(x).$$

Iterating yields $H_n^{(n)} = \mathcal{F}$. Every derivative of H_n below order n is a constant multiple of a scaled value of $\mathcal{F}$, hence vanishes at zero because $\mathcal{F}(0) = 0$. Uniqueness with its zero initial jet proves (15). If $x \leq 1$, the oriented interval between zero and x lies in $(-\infty, 1]$, where $F = \mathcal{F}$; moreover $x/2^n \leq 1$. Replacing $\mathcal{F}$ by F on both sides therefore proves (14) on the stated full range. $\square$

Remark 3.2. The range restriction $x \leq 1$ in (14) is the full interval on which the bounded function is obtained directly from the signed extension. It is $\mathcal{F}$, not the bounded CDF, that obeys the clean dilation equation on all of $\mathbb{R}$. The proposed global bounded completion is instead piecewise polynomial; see [theorem 5.1](#) and its frontier-status note.

Corollary 3.3 (Moment interpretation of the ladder). *For $n \geq 1$ and $x \geq 0$,*

$$(\mathcal{J}_n F)(x) = \frac{1}{n!} \mathbb{E}(x - X)_+^n. \quad (16)$$

In particular,

$$(\mathcal{J}_n F)(1) = \frac{d_n}{n!} = 2^{\binom{n}{2}} F(2^{-n}). \quad (17)$$

Proof. Write $F(t) = \mathbb{E}1_{\{X \leq t\}}$, apply Tonelli's theorem, and integrate $(x - t)^{n-1}$ from $t = X$ to $t = x$. At $x = 1$, symmetry gives $\mathbb{E}(1 - X)^n = \mathbb{E}X^n = d_n$. $\square$

4 The Newton–Volterra expansion for arbitrary powers

4.1 A general commutator identity

The key algebraic mechanism does not depend on the Fabius function.

Theorem 4.1 (Formalized polynomial Volterra commutator). *Let E be a real normed vector space, let $a, x \in \mathbb{R}$, let $f : \mathbb{R} \rightarrow E$ be interval integrable between a and x , and let $P \in \mathbb{R}[X]$. For every $n \in \mathbb{N}_0$,*

$$\mathcal{J}_{n,a}[P(t)f(t)](x) = \sum_{r=0}^{\deg_N P} (-1)^r n^{\bar{r}} P^{[r]}(x) \mathcal{J}_{n+r,a} f(x). \quad (18)$$

Here $P^{[r]}$ is the r -th Hasse derivative, $\deg_N$ is the natural degree, and $n^{\bar{r}}$ is the rising factorial, with $0^{\bar{0}} = 1$ and $0^{\bar{r}} = 0$ for $r > 0$. Consequently (18) includes the order-zero identity without a separate exception.

Proof. Taylor’s finite polynomial identity about the upper endpoint is

$$P(t) = \sum_{r=0}^{\deg_N P} P^{[r]}(x)(t-x)^r = \sum_{r=0}^{\deg_N P} (-1)^r P^{[r]}(x)(x-t)^r.$$

Linearity reduces the claim to the kernel-raising identity

$$\mathcal{J}_{n,a}[(x-t)^r f(t)](x) = n^{\bar{r}} \mathcal{J}_{n+r,a} f(x).$$

For positive n , this follows by combining the powers of $x-t$ and the factorial identity $(n-1)!n^{\bar{r}} = (n+r-1)!$. For $n=0$, evaluation at $t=x$ agrees with the stated zero rising-factorial convention. This proves all orders at once. $\square$

This theorem is exactly `normalizedVolterra_polynomial`; its monomial specialization is `normalizedVolterra_monomial`, both in `FabiusFunction.NormalizedVolterra`. Taking $P(t) = t^p$ gives $P^{[r]}(x) = \binom{p}{r} x^{p-r}$, so for $n \geq 1$ it is equivalent to the finite part of the next theorem.

Theorem 4.2 (Newton–Volterra commutator). *Let $n \geq 1$, let $x > 0$, and suppose the displayed integrals converge. Then for every complex α ,*

$$\mathcal{J}_n[t^\alpha f(t)](x) = \sum_{r=0}^{\infty} (-1)^r \binom{n+r-1}{r} \alpha^r x^{\alpha-r} (\mathcal{J}_{n+r} f)(x), \quad (19)$$

whenever termwise integration is justified. Independently of this positivity condition, if $\alpha = p \in \mathbb{N}_0$, the series terminates at $r = p$ and is an algebraic identity for every locally integrable f and every real oriented endpoint x :

$$\mathcal{J}_n[t^p f(t)](x) = \sum_{r=0}^p (-1)^r \binom{n+r-1}{r} \frac{p!}{(p-r)!} x^{p-r} (\mathcal{J}_{n+r} f)(x). \quad (20)$$

Proof. Inside the Volterra kernel, expand about the upper endpoint:

$$t^\alpha = x^\alpha \left(1 - \frac{x-t}{x}\right)^\alpha = \sum_{r \geq 0} (-1)^r \frac{\alpha^{\bar{r}}}{r!} x^{\alpha-r} (x-t)^r.$$

After integration, the factor multiplying $\mathcal{J}_{n+r} f$ is

$$\frac{\Gamma(n+r)}{\Gamma(n)r!} = \binom{n+r-1}{r}.$$

For $p \in \mathbb{N}_0$, the binomial expansion is finite, so no convergence argument is needed. $\square$

Formalization boundary for the commutator

The finite polynomial identity (18), including its natural-monomial specialization (20), is formalized in the greater normed-space-valued, arbitrary-base, all-order generality stated above. The infinite expansion (19) for nonintegral or complex α , including the justification of termwise integration, is not yet formalized.

For $p \geq 1$, the first two operator recurrences encoded in (20) are

$$\mathcal{J}_n[t f(t)] = x \mathcal{J}_n f - n \mathcal{J}_{n+1} f, \quad (21)$$

$$\mathcal{J}_n[t^p f(t)] = x \mathcal{J}_n[t^{p-1} f(t)] - n \mathcal{J}_{n+1}[t^{p-1} f(t)]. \quad (22)$$

They provide a convenient triangular implementation without symbolic integration.

4.2 The complete Fabius formula

Theorem 4.3 (Arbitrary-complex-power Fabius primitive). *Let $n \geq 1$, $0 < x \leq 1$, and $\alpha \in \mathbb{C}$. Then the integral $\mathcal{J}_n[t^\alpha F(t)](x)$ exists, is entire as a function of α , and has the absolutely and locally uniformly convergent expansion*

$$\mathcal{J}_n[t^\alpha F(t)](x) = \sum_{r=0}^{\infty} (-1)^r \binom{n+r-1}{r} \alpha^r 2^{\binom{n+r}{2}} x^{\alpha-r} F\left(\frac{x}{2^{n+r}}\right). \quad (23)$$

For $\alpha = p \in \mathbb{N}_0$, this terminates and becomes

$$\mathcal{J}_n[t^p F(t)](x) = \sum_{r=0}^p (-1)^r \binom{n+r-1}{r} \frac{p!}{(p-r)!} 2^{\binom{n+r}{2}} x^{p-r} F\left(\frac{x}{2^{n+r}}\right). \quad (24)$$

The natural-power specialization has the stronger all-order form: for every $n, p \in \mathbb{N}_0$ and every real $x \leq 1$,

$$\mathcal{J}_n[t^p F(t)](x) = \sum_{r=0}^p (-1)^r \binom{p}{r} n^{\bar{r}} x^{p-r} 2^{\binom{n+r}{2}} F\left(\frac{x}{2^{n+r}}\right). \quad (25)$$

For positive n , this is (24); the rising-factorial form also handles $n = 0$ without an exceptional sum.

Proof. Substitution of (14) into (19) gives the formal expression. It remains to justify convergence for all α .

Infinite flatness implies that for every integer $M \geq 0$ there is a finite constant C_M such that $F(t) \leq C_M t^M$ on $[0, 1]$. Hence, for every integer $m \geq 1$,

$$0 \leq (\mathcal{J}_m F)(x) \leq \frac{C_M M!}{(m+M)!} x^{m+M}. \quad (26)$$

Indeed, this is the beta integral obtained by replacing $F(t)$ with $C_M t^M$. The absolute value of the r -th term of (19), with $m = n + r$, is therefore at most

$$C_M M! x^{\operatorname{Re} \alpha + n + M} \binom{n+r-1}{r} \frac{|\alpha^r|}{(n+r+M)!}. \quad (27)$$

On a compact set of α 's, the falling factorial is bounded by $C r! r^B$ for some fixed B . Since $\binom{n+r-1}{r} = O(r^{n-1})$, choosing $M > B + 1$ makes (27) summable uniformly on that compact set. This proves absolute local uniform convergence. The same flatness bound, with logarithmic powers inserted, permits differentiation under the defining integral and shows that it is entire in α . The series and integral agree first in a half-plane where the ordinary binomial argument is immediate and then everywhere by the identity theorem; equivalently, one may use dominated convergence directly with (27).

For natural p , apply the finite monomial commutator (18) and then the bounded ladder (14). This argument has no convergence step and remains valid for every oriented endpoint $x \leq 1$ and for order zero, proving (25). $\square$

Formalization boundary for the Fabius formula

The finite identities (24) and (25) are formalized as `normalizedVolterra_pow_mul_fabiusReal_of_le_one`; the signed global version is `normalizedVolterra_pow_mul_extendedFabius`, both in `FabiusFunction.FabiusAntiderivatives`. The arbitrary-complex-power series (23), its entirety assertion, and its normal convergence are still non-formalized.

Main closed form for the ordinary antiderivative

For every real $x \leq 1$ and $p \in \mathbb{N}_0$, the normalized ordinary primitive is

$$\int_0^x t^p F(t) dt = \sum_{r=0}^p (-1)^r \frac{p!}{(p-r)!} 2^{\binom{r+1}{2}} x^{p-r} F\left(\frac{x}{2^{r+1}}\right). \quad (28)$$

Thus only $p + 1$ dyadically contracted Fabius evaluations are required.

A useful one-step recurrence for $p \geq 1$ and $x \leq 1$ follows by integrating by parts once and changing variables:

$$A_p(x) := \int_0^x t^p F(t) dt = x^p F(x/2) - p 2^p A_{p-1}(x/2), \quad A_0(x) = F(x/2). \quad (29)$$

Iteration of (29) is exactly (28).

4.3 Remainders and the negative-power series

For the ordinary primitive $n = 1$, repeated integration by parts gives an especially transparent exact remainder. For every integer $N \geq 0$,

$$\int_0^x t^\alpha F(t) dt = \sum_{r=0}^N (-1)^r \alpha^r x^{\alpha-r} (\mathcal{J}_{r+1} F)(x) + R_N(\alpha, x), \quad (30)$$

$$R_N(\alpha, x) = (-1)^{N+1} \alpha^{N+1} \int_0^x t^{\alpha-N-1} (\mathcal{J}_{N+1} F)(t) dt. \quad (31)$$

If $M > -\operatorname{Re} \alpha - 1$, (26) yields

$$|R_N(\alpha, x)| \leq \frac{|\alpha^{N+1}| C_M M!}{(N+M+1)! (\operatorname{Re} \alpha + M + 1)} x^{\operatorname{Re} \alpha + M + 1}. \quad (32)$$

This estimate is useful even when direct evaluation of the extremely small Fabius terms would underflow.

For a negative integer exponent $\alpha = -m$, $m \geq 1$, the signs cancel and every term is positive:

$$\int_0^x t^{-m} F(t) dt = \sum_{r=0}^{\infty} \frac{(m+r-1)!}{(m-1)!} 2^{\binom{r+1}{2}} x^{-m-r} F\left(\frac{x}{2^{r+1}}\right). \quad (33)$$

No principal value or regularization is involved: the flat zero of F makes $x^{-m} F(x)$ integrable for every m . This behavior is sharply different from $x^{-m} \operatorname{up}(x)$, which is singular at the origin because $\operatorname{up}(0) = 1$.

Corollary 4.4 (Endpoint localization). *Fix $n \geq 1$ and $\alpha \in \mathbb{C}$ with $\operatorname{Re} \alpha \geq 0$. As $x \downarrow 0$,*

$$\mathcal{J}_n[t^\alpha F(t)](x) \sim x^\alpha (\mathcal{J}_n F)(x) = x^\alpha 2^{\binom{n}{2}} F(x/2^n). \quad (34)$$

Proof. The sharp endpoint theory in the audited corpus implies rapid variation: for every fixed $0 < c < 1$, $F(cx)/F(x) \rightarrow 0$ as $x \downarrow 0$. Normalize the nonnegative measure $(x-t)^{n-1} F(t) dt$ on $[0, x]$. Its mass below $(1-\delta)x$ is negligible relative to its total mass, because comparison with fixed dilates of F shows concentration in $[(1-\delta)x, x]$. On that interval $(t/x)^\alpha \rightarrow 1$ uniformly as $\delta \downarrow 0$. Moreover, $|(t/x)^\alpha| = (t/x)^{\operatorname{Re} \alpha} \leq 1$ on $(0, x]$, so the complementary part contributes at most twice its vanishing normalized mass. The quotient of the two sides of (34) is the expectation of $(t/x)^\alpha$ under this normalized measure, hence tends to one. $\square$

For $\operatorname{Re} \alpha < 0$, the same localization is expected and is consistent with the exact Newton–Volterra series, but the preceding concentration argument alone does not prove it: $(t/x)^\alpha$ is then unbounded near zero. Extending (34) to that half-plane requires a separate uniform-integrability estimate from quantitative flatness (or from the sharp endpoint expansion), and no such extension is asserted here.

The first correction displayed by (23) is

$$\frac{\mathcal{J}_n[t^\alpha F(t)](x)}{x^\alpha \mathcal{J}_n F(x)} = 1 - n\alpha \frac{\mathcal{J}_{n+1} F(x)}{x \mathcal{J}_n F(x)} + \dots \quad (35)$$

The quotient on the right is governed by the repository’s Lambert phase and periodic saddle fluctuation; extracting its complete expansion is posed as a frontier problem in [section 16](#).

5 Global closed form for natural powers

For integer powers the bounded extension of F permits a complete all-real answer. Define

$$H_{n,p}(x) = \frac{1}{(n-1)!} \int_0^x (x-t)^{n-1} t^p F(t) dt, \quad (36)$$

with the usual oriented integral when $x < 0$.

Formalization frontier

The finite dyadic branch for every $x \leq 1$, including the zero branch on the negative half-line, follows from the formalized theorem `normalizedVolterra_pow_mul_fabiusReal_of_le_one`. The exterior moment-polynomial branch for $x > 1$, the resulting all-real piecewise assembly, and the matching-jet assertion at $x = 1$ have not yet been formalized. Thus [theorem 5.1](#) remains in this frontier report.

Theorem 5.1 (Global piecewise formula). *For $n \geq 1$ and $p \in \mathbb{N}_0$,*

$$H_{n,p}(x) = \begin{cases} 0, & x \leq 0, \\ \sum_{r=0}^p (-1)^r \binom{n+r-1}{r} \frac{p!}{(p-r)!} 2^{\binom{n+r}{2}} x^{p-r} F\left(\frac{x}{2^{n+r}}\right), & 0 \leq x \leq 1, \\ \frac{p!}{(p+n)!} x^{p+n} - \frac{1}{(n-1)!} \sum_{j=0}^{n-1} (-1)^j \binom{n-1}{j} x^{n-1-j} \frac{d_{p+j+1}}{p+j+1}, & x \geq 1. \end{cases} \quad (37)$$

The pieces have matching derivatives through order $n-1$ at zero and one.

Proof. The first two cases follow from the support convention and [theorem 4.3](#). For $x \geq 1$, write

$$H_{n,p}(x) = \mathcal{J}_n[t^p](x) - \frac{1}{(n-1)!} \int_0^1 (x-t)^{n-1} t^p (1-F(t)) dt.$$

The first term is $p!x^{p+n}/(p+n)!$. Expand the finite polynomial kernel and use

$$\int_0^1 t^q (1-F(t)) dt = \frac{d_{q+1}}{q+1}, \quad (38)$$

which is the standard tail integral for $\mathbb{E}X^{q+1}$. Since the formula was derived from a single Volterra integral, matching jets are automatic. $\square$

For the ordinary primitive, the last branch simplifies to

$$A_p(x) = \frac{x^{p+1} - d_{p+1}}{p+1}, \quad x \geq 1. \quad (39)$$

This makes clear that the normalized antiderivative is constant-polynomial outside the transition interval, exactly as repeated integration of a compact transition requires.

6 Definite integrals, Newton interpolation, and moment identities

6.1 The entire Mellin moment

Endpoint flatness makes

$$M(s) = \mathbb{E}X^s = \int_0^1 x^s \, dF(x) \quad (40)$$

an entire function of $s \in \mathbb{C}$. Integration by parts, initially in a right half-plane and then by analytic continuation, gives

$$\mathcal{A}(\alpha) := \int_0^1 x^\alpha F(x) \, dx = \frac{1 - M(\alpha + 1)}{\alpha + 1}. \quad (41)$$

The apparent singularity at $\alpha = -1$ is removable.

Putting $x = 1$ in (23) and using (17) gives the central Newton series.

Theorem 6.1 (Moment Newton series). *For every $\alpha \in \mathbb{C}$,*

$$\mathcal{A}(\alpha) = \sum_{r=0}^{\infty} (-1)^r \binom{\alpha}{r} \frac{d_{r+1}}{r+1}. \quad (42)$$

More generally, for every integer $n \geq 1$,

$$\mathcal{J}_n[t^\alpha F(t)](1) = \frac{1}{(n-1)!} \sum_{r=0}^{\infty} (-1)^r \binom{\alpha}{r} \frac{d_{n+r}}{n+r}. \quad (43)$$

Both series converge absolutely and locally uniformly in α .

The terminology is literal. If $\Delta h(z) = h(z+1) - h(z)$, then

$$\Delta^r \mathcal{A}(0) = \int_0^1 (x-1)^r F(x) \, dx = (-1)^r \frac{d_{r+1}}{r+1}. \quad (44)$$

Thus (42) is the Newton interpolation series of the entire function $\mathcal{A}$ from its nonnegative integral values.

For $p \in \mathbb{N}_0$, the series terminates and combines with (41) to give

$$\int_0^1 x^p F(x) \, dx = \frac{1 - d_{p+1}}{p+1}. \quad (45)$$

Together with (8), this yields the exact finite dyadic identity

$$\sum_{r=0}^p (-1)^r \frac{p!}{(p-r)!} 2^{\binom{r+1}{2}} F(2^{-r-1}) = \frac{1 - d_{p+1}}{p+1}. \quad (46)$$

Every value in (46) is rational and may be replaced by the repository's explicit q -binomial or integer-sequence formulas.

6.2 A two-parameter finite-difference identity

Expanding $(1 - t)^{n-1}$ in the defining integral at $x = 1$ produces an independent form. Equating it with (43) gives the following exact identity.

Corollary 6.2 (Two-parameter moment identity). *For $n \geq 1$ and $p \in \mathbb{N}_0$,*

$$\sum_{r=0}^p (-1)^r \binom{p}{r} \frac{d_{n+r}}{n+r} = \sum_{j=0}^{n-1} (-1)^j \binom{n-1}{j} \frac{1 - d_{p+j+1}}{p+j+1}. \quad (47)$$

More generally, for every $\alpha \in \mathbb{C}$, the finite left-hand side is replaced by its convergent Newton series and d_{p+j+1} by $M(\alpha + j + 1)$:

$$\sum_{r \geq 0} (-1)^r \binom{\alpha}{r} \frac{d_{n+r}}{n+r} = \sum_{j=0}^{n-1} (-1)^j \binom{n-1}{j} \frac{1 - M(\alpha + j + 1)}{\alpha + j + 1}, \quad (48)$$

where removable values are understood.

6.3 Negative powers and logarithmic constants

For $m \geq 1$, (42) becomes the positive series

$$\int_0^1 x^{-m} F(x) dx = \sum_{n \geq 1} \frac{(m+n-2)!}{(m-1)! n!} d_n. \quad (49)$$

For $m > 1$, the Mellin form is

$$\int_0^1 x^{-m} F(x) dx = \frac{M(1-m) - 1}{m-1}. \quad (50)$$

At the removable case $m = 1$,

$$\int_0^1 \frac{F(x)}{x} dx = -M'(0) = -\mathbb{E} \log X = \sum_{n \geq 1} \frac{d_n}{n}. \quad (51)$$

The centered moments give a second positive expansion. Since

$$M(s) = 2^{-s} \sum_{q \geq 0} \binom{s}{2q} c_q, \quad (52)$$

and $\left. \frac{d}{ds} \binom{s}{2q} \right|_{s=0} = -1/(2q)$, one obtains

$$\int_0^1 \frac{F(x)}{x} dx = \log 2 + \frac{1}{2} \sum_{q \geq 1} \frac{c_q}{q}. \quad (53)$$

Numerically, the accompanying code gives

$$\int_0^1 \frac{F(x)}{x} dx = 0.75824000040440865 \dots \quad (54)$$

7 Exponential master functions and the sinc product

The entire family of integer-power primitives is encoded by an incomplete Laplace transform. Applying [theorem 4.2](#) to e^{zt} , or simply expanding $e^{-z(x-t)}$ in the Volterra kernel, gives the following.

Theorem 7.1 (Exponential Volterra master). *For $n \geq 1$, $0 \leq x \leq 1$, and $z \in \mathbb{C}$,*

$$\mathcal{J}_n[e^{zt}F(t)](x) = e^{zx} \sum_{r=0}^{\infty} (-z)^r \binom{n+r-1}{r} 2^{\binom{n+r}{2}} F\left(\frac{x}{2^{n+r}}\right). \quad (55)$$

The series is entire in z . At ordinary order $n = 1$,

$$\int_0^x e^{zt} F(t) dt = e^{zx} \sum_{m \geq 1} (-z)^{m-1} 2^{\binom{m}{2}} F(x/2^m). \quad (56)$$

At $x = 1$,

$$\int_0^1 e^{zt} F(t) dt = \frac{e^z - G(z)}{z} = e^z \frac{1 - G(-z)}{z}, \quad (57)$$

with the removable value $1/2$ at $z = 0$.

Taking p derivatives with respect to z at zero recovers [\(24\)](#). Thus [\(55\)](#) is a compact generating object for all natural monomial weights and all integral antiderivative orders.

For the up-function, define the incomplete transform

$$U(z, x) = \int_{-1}^x e^{zt} \text{up}(t) dt. \quad (58)$$

On the left half, translation gives the particularly simple dyadic series

$$U(z, x) = e^{zx} \sum_{m \geq 1} (-z)^{m-1} 2^{\binom{m}{2}} F\left(\frac{x+1}{2^m}\right), \quad -1 \leq x \leq 0. \quad (59)$$

At the right endpoint,

$$\int_{-1}^1 e^{zt} \text{up}(t) dt = e^{-z} G(2z) = \frac{G(z) - G(-z)}{z}. \quad (60)$$

For $z = 2\pi i\xi$, this is precisely the infinite sinc product, up to the chosen Fourier normalization. Consequently all compact up-moments and their incomplete analogues arise by differentiating one dyadic Fabius series whose complete endpoint is the sinc product.

8 Bell and Bernoulli polynomial organization

Taking logarithms of [\(10\)](#) and using the classical Bernoulli expansion gives

$$\log G(z) = \frac{z}{2} + \sum_{q \geq 1} \frac{B_{2q}}{2q(2q)!(2^{2q} - 1)} z^{2q}. \quad (61)$$

Hence the cumulants of X are

$$\kappa_1 = \frac{1}{2}, \quad \kappa_{2q} = \frac{B_{2q}}{2q(2^{2q} - 1)}, \quad \kappa_{2q+1} = 0 \quad (q \geq 1). \quad (62)$$

Writing Y_n for the complete exponential Bell polynomial,

$$d_n = Y_n(\kappa_1, \dots, \kappa_n). \quad (63)$$

Therefore every definite natural-power primitive has the explicit Bell–Bernoulli form

$$\int_0^1 x^p F(x) dx = \frac{1 - Y_{p+1}(\kappa_1, \dots, \kappa_{p+1})}{p+1}. \quad (64)$$

Substitution in (37) similarly expresses every exterior polynomial coefficient through Bernoulli numbers and Bell polynomials. This provides a direct bridge between the antiderivative problem and the Bell–Bernoulli infrastructure already emphasized in the repository.

9 Reflection transforms: $F(1-x)$ and $1-F(x)$

The global reflection identity (1) means that the two transformations requested in the problem are the same function:

$$F(1-x) = 1-F(x), \quad x \in \mathbb{R}. \quad (65)$$

For natural powers and every integer $n \geq 1$, linearity gives the complete normalized formula

$$\mathcal{J}_n[x^p(1-F(x))] = \frac{p!}{(p+n)!} x^{p+n} - H_{n,p}(x), \quad (66)$$

where $H_{n,p}$ is the global piecewise expression (37). The same formula therefore applies verbatim to $x^p F(1-x)$.

For arbitrary complex α on $x > 0$, an ordinary antiderivative is

$$\int x^\alpha (1-F(x)) dx = \begin{cases} \frac{x^{\alpha+1}}{\alpha+1} - \int_0^x t^\alpha F(t) dt + C, & \alpha \neq -1, \\ \log x - \int_0^x \frac{F(t)}{t} dt + C, & \alpha = -1, \end{cases} \quad (67)$$

where the Fabius term is evaluated by (23). Unlike the Fabius integral, a primitive normalized at zero exists only when $\operatorname{Re} \alpha > -1$, because $1-F(0) = 1$. Formula (67) remains a valid local antiderivative on every positive interval avoiding zero for all α .

At the complete endpoint, for $\operatorname{Re} \alpha > -1$,

$$\int_0^1 x^\alpha (1-F(x)) dx = \frac{M(\alpha+1)}{\alpha+1}. \quad (68)$$

Together, (41) and (68) split the elementary beta integral $1/(\alpha+1)$ into the lower-CDF and upper-tail contributions.

10 Rvachev up-function: complete weighted primitives

10.1 First-order formulas and the singularity at zero

On $0 \leq x \leq 1$, $\operatorname{up}(x) = 1-F(x)$, so for $\operatorname{Re} \alpha > -1$,

$$\int_0^x t^\alpha \operatorname{up}(t) dt = \frac{x^{\alpha+1}}{\alpha+1} - \sum_{r \geq 0} (-1)^r \alpha^r 2^{\binom{r+1}{2}} x^{\alpha-r} F\left(\frac{x}{2^{r+1}}\right). \quad (69)$$

The sum is the convergent Fabius series from (23). For $\alpha = -1$, replace the elementary term by $\log x$, obtaining a local primitive but not a convergent integral from zero.

This exposes an important structural contrast:

- $x^\alpha F(x)$ is integrable at zero for every $\alpha \in \mathbb{C}$, because F is infinitely flat;
- $x^\alpha \text{up}(x)$ has the ordinary monomial threshold $\text{Re } \alpha > -1$, because $\text{up}(0) = 1$.

For a chosen upper-half-plane branch on the negative axis, $(-u)^\alpha = e^{\pi i \alpha} u^\alpha$, evenness gives, whenever the integrals converge,

$$\int_{-1}^x t^\alpha \text{up}(t) dt = e^{\pi i \alpha} \left[\int_0^1 u^\alpha \text{up}(u) du - \int_0^{-x} u^\alpha \text{up}(u) du \right], \quad -1 \leq x \leq 0. \quad (70)$$

For integer powers this becomes real and simplifies to finite Fabius sums.

10.2 All natural powers and all integer orders

Define the primitive based at the left support endpoint,

$$U_{n,p}(x) = \frac{1}{(n-1)!} \int_{-1}^x (x-t)^{n-1} t^p \text{up}(t) dt, \quad n \geq 1, p \in \mathbb{N}_0. \quad (71)$$

Let $H_{n,j}$ be as in (36), and define the full up-moments

$$m_k^{\text{up}} = \int_{-1}^1 t^k \text{up}(t) dt = \begin{cases} 0, & k \text{ odd}, \\ c_{k/2}, & k \text{ even}. \end{cases} \quad (72)$$

Theorem 10.1 (Global weighted up-primitive). *For $n \geq 1$ and $p \in \mathbb{N}_0$,*

$$U_{n,p}(x) = \begin{cases} 0, & x \leq -1, \\ \sum_{j=0}^p \binom{p}{j} (-1)^{p-j} H_{n,j}(x+1), & -1 \leq x \leq 0, \\ \frac{(-1)^p}{(n-1)!} \sum_{j=0}^{n-1} \binom{n-1}{j} x^{n-1-j} \frac{d_{p+j+1}}{p+j+1} + \frac{p!}{(p+n)!} x^{p+n} - H_{n,p}(x), & 0 \leq x \leq 1, \\ \frac{1}{(n-1)!} \sum_{j=0}^{n-1} (-1)^j \binom{n-1}{j} x^{n-1-j} m_{p+j}^{\text{up}}, & x \geq 1. \end{cases} \quad (73)$$

Every branch is finite, and the interior branches reduce further to the scaled-Fabius formula (24).

Proof. For $-1 \leq x \leq 0$, set $s = t + 1$. Then $\text{up}(t) = F(s)$, $t^p = (s-1)^p$, and the first finite sum follows. For $0 \leq x \leq 1$, split the integral at zero. The negative half is

$$\frac{(-1)^p}{(n-1)!} \int_0^1 (x+1-s)^{n-1} (1-s)^p F(s) ds.$$

Expand the polynomial in x and use

$$\int_0^1 (1-s)^q F(s) ds = \frac{d_{q+1}}{q+1},$$

which follows from [corollary 3.3](#). The positive half equals $\mathcal{J}_n[t^p] - H_{n,p}$. For $x \geq 1$, the full support has been crossed; expand $(x-t)^{n-1}$ and insert the moments (72). $\square$

The unweighted case is worth displaying separately. Put $U_n = U_{n,0}$. Then

$$U_n(x) = \begin{cases} 0, & x \leq -1, \\ 2^{\binom{n}{2}} F\left(\frac{x+1}{2^n}\right), & -1 \leq x \leq 0, \\ \frac{1}{(n-1)!} \sum_{j=0}^{n-1} \binom{n-1}{j} \frac{d_{j+1}}{j+1} x^{n-1-j} + \frac{x^n}{n!} - 2^{\binom{n}{2}} F\left(\frac{x}{2^n}\right), & 0 \leq x \leq 1, \\ \frac{1}{(n-1)!} \sum_{q=0}^{\lfloor (n-1)/2 \rfloor} \binom{n-1}{2q} c_q x^{n-1-2q}, & x \geq 1. \end{cases} \quad (74)$$

For $n = 1$, this is the cumulative distribution function of $Y = 2X - 1$. For larger n , its right tail is the moment polynomial of the compact up density.

11 Exact higher derivatives and inverse differentiation

11.1 Thue–Morse derivative formulas

Let $\varepsilon_k = (-1)^{s_2(k)}$. Iterating Rvachev's functional-differential equation gives the standard exact derivative formula

$$\text{up}^{(n)}(x) = 2^{\binom{n+1}{2}} \sum_{k=0}^{2^n-1} \varepsilon_k \text{up}(2^n x + 2^n - 1 - 2k). \quad (75)$$

Using $F'(x) = 2 \text{up}(2x - 1)$, one obtains, for $m \geq 1$,

$$F^{(m)}(x) = 2^{\binom{m+1}{2}} \sum_{k=0}^{2^{m-1}-1} \varepsilon_k \text{up}(2^m x - 1 - 2k). \quad (76)$$

This version is valid globally for the bounded F . On the shrinking left cell $0 < x < 2^{-m}$, it collapses to

$$F^{(m)}(x) = 2^{\binom{m+1}{2}} F(2^m x), \quad (77)$$

while the signed extension satisfies the same dilation formula on all of $\mathbb{R}$:

$$\mathcal{F}^{(m)}(x) = 2^{\binom{m+1}{2}} \mathcal{F}(2^m x). \quad (78)$$

Leibniz's rule therefore gives a closed formula for the requested derivatives. For $p \in \mathbb{N}_0$,

$$D^n[x^p \mathcal{F}(x)] = \sum_{j=0}^{\min(n,p)} \binom{n}{j} \frac{p!}{(p-j)!} x^{p-j} 2^{\binom{n-j+1}{2}} \mathcal{F}(2^{n-j} x). \quad (79)$$

For arbitrary $\alpha \in \mathbb{C}$ and $x > 0$, replace $p!/(p-j)!$ by α^j and sum through $j = n$. For the bounded function, use

$$D^n[x^\alpha F(x)] = \sum_{j=0}^n \binom{n}{j} \alpha^j x^{\alpha-j} F^{(n-j)}(x), \quad (80)$$

with (76). An identical Leibniz formula combined with (75) handles $x^\alpha \text{up}(x)$.

11.2 The derivative/antiderivative dictionary

The formulas may be summarized as follows.

- Positive derivative order introduces a finite Thue–Morse superposition of dilated up-functions, or a single dilation for $\mathcal{F}$.
- Negative integral order on $[0, 1]$ introduces a single contracted Fabius value, multiplied by $2^{\binom{n}{2}}$.
- Multiplication by a natural monomial turns either operation into a finite triangular sum; a nonintegral monomial changes the antiderivative sum from finite to absolutely convergent.

This is the exact operator closure that replaces an elementary integration table for the Fabius family.

12 Fractional antiderivative order

The integer ladder has a natural Riemann–Liouville continuation, although the single-dilation collapse is special to integral order.

Proposition 12.1 (Probabilistic fractional primitive). *For $\operatorname{Re} \nu > 0$ and $x \geq 0$,*

$$(\mathcal{I}^\nu F)(x) = \frac{1}{\Gamma(\nu + 1)} \mathbb{E}(x - X)_+^\nu. \quad (81)$$

At $x = 1$, symmetry gives

$$(\mathcal{I}^\nu F)(1) = \frac{M(\nu)}{\Gamma(\nu + 1)}. \quad (82)$$

For $x > 1$,

$$(\mathcal{I}^\nu F)(x) = \frac{x^\nu}{\Gamma(\nu + 1)} \sum_{k \geq 0} (-1)^k \binom{\nu}{k} d_k x^{-k}; \quad (83)$$

for $x = 1$, endpoint flatness supplies the limiting interpretation.

Proof. Since $\operatorname{Re} \nu > 0$, the absolute value of the kernel is $(x - t)^{\operatorname{Re} \nu - 1}$, which is integrable on the finite triangular region arising after insertion of $F(t) = \mathbb{E} \mathbf{1}_{\{X \leq t\}}$. Fubini's theorem therefore gives (81); for positive real ν , this reduces to the Tonelli argument in [corollary 3.3](#). At one, replace $1 - X$ by the identically distributed X . For $x > 1$, expand $(1 - X/x)^\nu$ and take expectations. $\square$

Theorem 12.2 (Fractional-order Newton–Volterra formula). *Let $\operatorname{Re} \nu > 0$, $0 < x \leq 1$, and $\alpha \in \mathbb{C}$. Then*

$$\mathcal{I}^\nu [t^\alpha F(t)](x) = \sum_{r \geq 0} (-1)^r \frac{\nu^{\bar{r}}}{r!} \alpha^r x^{\alpha - r} (\mathcal{I}^{\nu+r} F)(x). \quad (84)$$

The series converges absolutely and locally uniformly in (α, ν) on compact subsets of $\mathbb{C} \times \{\operatorname{Re} \nu > 0\}$. For $p \in \mathbb{N}_0$, it terminates:

$$\mathcal{I}^\nu [t^p F(t)](x) = \sum_{r=0}^p (-1)^r \binom{\nu + r - 1}{r} \frac{p!}{(p-r)!} x^{p-r} (\mathcal{I}^{\nu+r} F)(x). \quad (85)$$

At $x = 1$,

$$\mathcal{I}^\nu [t^\alpha F(t)](1) = \frac{1}{\Gamma(\nu)} \sum_{r \geq 0} (-1)^r \binom{\alpha}{r} \frac{M(\nu + r)}{\nu + r}. \quad (86)$$

Proof. Expand t^α about x exactly as in [theorem 4.2](#). Raising the kernel power from $\nu - 1$ to $\nu + r - 1$ contributes $\Gamma(\nu + r)/\Gamma(\nu) = \nu^{\bar{r}}$. The flatness estimate used in [theorem 4.3](#), with gamma functions replacing factorials, proves normal convergence. Formula (86) follows from (82) and

$$\frac{\nu^{\bar{r}}}{r!} \alpha^r \frac{1}{\Gamma(\nu + r + 1)} = \frac{1}{\Gamma(\nu)} \binom{\alpha}{r} \frac{1}{\nu + r}.$$

□

This fractional identity is an exact analytic continuation of the complete integer-order table. It also shows that the moment function $M(s)$, rather than a new special function, is the natural endpoint datum for fractional Fabius integration.

13 Antiderivatives of the inverse Fabius function

13.1 Exact first-order reductions

Let

$$Q(y) = F^{-1}(y), \quad 0 \leq y \leq 1. \quad (87)$$

The repository establishes strict monotonicity of F , smoothness of Q on $(0, 1)$, reflection

$$Q(1 - y) = 1 - Q(y), \quad (88)$$

and infinite endpoint slope [1, 2]. Substitution $u = F(t)$ yields an unexpectedly compact primitive.

Theorem 13.1 (Closed primitive of the inverse). *For $0 \leq y \leq 1$,*

$$\boxed{\int_0^y Q(u) \, du = yQ(y) - F\left(\frac{Q(y)}{2}\right)}. \quad (89)$$

In particular,

$$\int_0^1 Q(u) \, du = \frac{1}{2}. \quad (90)$$

Proof. Put $x = Q(y)$. Then

$$\int_0^y Q(u) \, du = \int_0^x tF'(t) \, dt = xF(x) - \int_0^x F(t) \, dt.$$

Apply $F(x) = y$ and the base ladder $\int_0^x F(t) \, dt = F(x/2)$. At $y = 1$, use $Q(1) = 1$ and $F(1/2) = 1/2$. □

For a complex weight α , the same substitution and one integration by parts give the following exact reduction to a power of F .

Proposition 13.2 (Weighted inverse primitive). *Let $0 < y \leq 1$ and $x = Q(y)$. If $\alpha \neq -1$ and $\operatorname{Re} \alpha \geq -1$, then*

$$\boxed{\int_0^y u^\alpha Q(u) \, du = \frac{y^{\alpha+1} Q(y)}{\alpha + 1} - \frac{1}{\alpha + 1} \int_0^{Q(y)} F(t)^{\alpha+1} \, dt.} \quad (91)$$

At the exceptional exponent,

$$\boxed{\int_0^y \frac{Q(u)}{u} \, du = Q(y) \log y - \int_0^{Q(y)} \log F(t) \, dt.} \quad (92)$$

For real $\alpha < -1$ and every $y > 0$, the improper integral diverges. At $y = 0$, the convergent cases above extend with value zero by taking the right-hand limit; this convention is needed because the displayed factors $y^{\alpha+1}$ and $\log y$ are not themselves defined at zero.

Proof. After $u = F(t)$, the left side is $\int_0^x t F(t)^\alpha F'(t) dt$. Use $F^\alpha F' = (F^{\alpha+1})'/(\alpha+1)$. On the boundary line $\operatorname{Re} \alpha = -1$, the endpoint factor still vanishes because of the multiplying t . At $\alpha = -1$, integrate $t(\log F(t))'$. The repository's endpoint asymptotic implies $t \log F(t) \rightarrow 0$. Finally, $Q(u)$ decays more slowly than every positive power of u , so $u^\alpha Q(u)$ cannot be integrable when $\alpha < -1$. $\square$

At $y = 1$ and $p \in \mathbb{N}_0$,

$$\int_0^1 u^p Q(u) du = \frac{1}{p+1} \left(1 - \int_0^1 F(t)^{p+1} dt \right). \quad (93)$$

If $X_1, \dots, X_{p+1}$ are independent Fabius variables, then

$$\int_0^1 u^p Q(u) du = \frac{1}{p+1} \mathbb{E} \max(X_1, \dots, X_{p+1}). \quad (94)$$

Thus weighted quantile integrals are normalized expected maxima. At $p = -1$, formula (92) gives the convergent constant

$$\int_0^1 \frac{Q(u)}{u} du = - \int_0^1 \log F(t) dt. \quad (95)$$

13.2 Higher and fractional inverse primitives

The indicator representation

$$Q(u) = \int_0^1 \mathbf{1}_{\{F(t) \leq u\}} dt \quad (96)$$

turns all higher primitives into one-dimensional integrals of powers of $y - F(t)$.

Theorem 13.3 (All orders for the inverse). *For $\operatorname{Re} \nu > 0$ and $0 \leq y \leq 1$,*

$$(\mathcal{I}^\nu Q)(y) = \frac{1}{\Gamma(\nu+1)} \int_0^{Q(y)} (y - F(t))^\nu dt. \quad (97)$$

For an integer $n \geq 1$, this is

$$(\mathcal{J}_n Q)(y) = \frac{1}{n!} \int_0^{Q(y)} (y - F(t))^n dt. \quad (98)$$

At the complete endpoint,

$$(\mathcal{I}^\nu Q)(1) = \frac{1}{\Gamma(\nu+1)} \int_0^1 F(t)^\nu dt, \quad (99)$$

where reflection was used in the last step.

Proof. Insert (96) into (13). Because $\operatorname{Re} \nu > 0$, the absolute value of the resulting kernel is integrable on the finite region $0 \leq F(t) \leq u \leq y$; hence Fubini's theorem permits interchanging the integrals. For a fixed $t \leq Q(y)$, the variable u ranges from $F(t)$ to y , and

$$\frac{1}{\Gamma(\nu)} \int_{F(t)}^y (y - u)^{\nu-1} du = \frac{(y - F(t))^\nu}{\Gamma(\nu+1)}.$$

At $y = 1$, substitute $t \mapsto 1 - t$ and use $1 - F(t) = F(1 - t)$. $\square$

For a natural weight p , the same argument gives

$$\mathcal{I}^\nu[u^p Q(u)](y) = \frac{1}{\Gamma(\nu)} \int_0^{Q(y)} \int_{F(t)}^y (y-u)^{\nu-1} u^p \, du \, dt. \quad (100)$$

For $y > 0$, the inner integral is an incomplete beta function:

$$y^{p+\nu} [B(p+1, \nu) - B_{F(t)/y}(p+1, \nu)]. \quad (101)$$

When p is a nonnegative integer and $\nu = n$ is a positive integer, it is a finite polynomial in y and $F(t)$.

13.3 Higher derivatives of the inverse

Let $x = Q(y) \in (0, 1)$. The classical Lagrange inverse-derivative formula [17] gives an exact all-orders identity that can be combined with the Thue–Morse derivative expression (76). We import that classical identity here rather than reprove Lagrange inversion.

Theorem 13.4 (Lagrange formula for inverse derivatives). *For $n \geq 1$,*

$$Q^{(n)}(y) = \frac{d^{n-1}}{dt^{n-1}} \left(\frac{t-x}{F(t)-F(x)} \right)^n \Big|_{t=x}. \quad (102)$$

If

$$a_m(x) = \frac{F^{(m+1)}(x)}{(m+1)!F'(x)}, \quad (103)$$

then equivalently

$$Q^{(n)}(y) = \frac{(n-1)!}{F'(x)^n} [z^{n-1}] \left(1 + \sum_{m \geq 1} a_m(x) z^m \right)^{-n}. \quad (104)$$

Indeed, writing $t = x + z$ gives

$$\frac{t-x}{F(t)-F(x)} = \frac{1}{F'(x)} \left(1 + \sum_{m \geq 1} a_m(x) z^m \right)^{-1}.$$

Raising this identity to the n -th power and taking the $(n-1)$ -st derivative at $z = 0$ multiplies the coefficient of z^{n-1} by $(n-1)!$, proving the equivalence of (102) and (104).

The first cases are

$$Q' = \frac{1}{F'}, \quad (105)$$

$$Q'' = -\frac{F''}{(F')^3}, \quad (106)$$

$$Q''' = \frac{3(F'')^2 - F'F'''}{(F')^5}, \quad (107)$$

$$Q'''' = \frac{-15(F'')^3 + 10F'F''F''' - (F')^2F''''}{(F')^7}, \quad (108)$$

all evaluated at $x = Q(y)$. Inserting (76) makes each one a finite Thue–Morse expression in translated, dilated up-values. On a left dyadic cell, (77) further collapses every derivative of F to a single scaled Fabius value.

14 New inverse-primitive asymptotics

14.1 Repository input and slow variation

The consolidated frontier volume derives from the sharp Lambert expansion the endpoint equivalent

$$Q(y) \sim \frac{\sqrt{\log(1/y)}}{e\sqrt{\log 2}} \exp\left[-\sqrt{2 \log 2 \log(1/y)}\right] \quad (y \downarrow 0). \quad (109)$$

At the audited snapshot this statement was marked “Derived” rather than Lean-formalized in the source. The current Lean development now proves precisely this equivalence as `Fabius.fabiusInv_isEquivalent_fabiusInverseAsymptoticMain` in `FabiusInverseAsymptotic.lean`. Thus the results below are unconditional mathematical consequences of a now-formalized input, although their integral and order-statistic deductions have not themselves yet been translated into Lean [2].

Put

$$L = \log 2, \quad a = \sqrt{2L}, \quad K = \frac{1}{e\sqrt{L}}. \quad (110)$$

Then $Q(y) \sim K\sqrt{T}e^{-a\sqrt{T}}$, where $T = \log(1/y)$. For every fixed $c > 0$, replacing T by $T - \log c$ changes this expression by a factor tending to one. Therefore

$$\frac{Q(cy)}{Q(y)} \rightarrow 1, \quad y \downarrow 0, \quad (111)$$

so Q is slowly varying at zero.

14.2 Weighted primitives near zero

Theorem 14.1 (Karamata law for inverse primitives). *Let $\alpha > -1$ and $\nu > 0$ be real. Then, as $y \downarrow 0$,*

$$\mathcal{I}^\nu[u^\alpha Q(u)](y) \sim \frac{\Gamma(\alpha+1)}{\Gamma(\alpha+\nu+1)} y^{\alpha+\nu} Q(y). \quad (112)$$

In particular,

$$\int_0^y u^\alpha Q(u) du \sim \frac{y^{\alpha+1} Q(y)}{\alpha+1}, \quad (113)$$

$$(\mathcal{I}^\nu Q)(y) \sim \frac{y^\nu Q(y)}{\Gamma(\nu+1)}. \quad (114)$$

Proof. Scale $u = ys$ in the defining integral:

$$\mathcal{I}^\nu[u^\alpha Q(u)](y) = \frac{y^{\alpha+\nu} Q(y)}{\Gamma(\nu)} \int_0^1 (1-s)^{\nu-1} s^\alpha \frac{Q(ys)}{Q(y)} ds.$$

Pointwise convergence to one follows from (111). Potter bounds for slowly varying functions provide an integrable dominating beta kernel. The limiting integral is $B(\alpha+1, \nu)/\Gamma(\nu) = \Gamma(\alpha+1)/\Gamma(\alpha+\nu+1)$. $\square$

The boundary exponent $\alpha = -1$ is convergent but belongs to a different scale.

Theorem 14.2 (Logarithmic inverse primitive). *As $y \downarrow 0$, with $T = \log(1/y)$,*

$$\int_0^y \frac{Q(u)}{u} du \sim \frac{2\sqrt{T}}{a} Q(y) \quad (115)$$

$$\sim \frac{\sqrt{2}}{e \log 2} T \exp\left[-\sqrt{2 \log 2 T}\right]. \quad (116)$$

Proof. With $u = e^{-s}$,

$$\int_0^y \frac{Q(u)}{u} du = \int_T^\infty Q(e^{-s}) ds.$$

The input equivalent turns the integrand into $K\sqrt{s}e^{-a\sqrt{s}}$. Substituting $v = \sqrt{s}$ gives

$$2K \int_{\sqrt{T}}^\infty v^2 e^{-av} dv = 2K e^{-a\sqrt{T}} \left(\frac{T}{a} + \frac{2\sqrt{T}}{a^2} + \frac{2}{a^3} \right).$$

The leading term is $(2\sqrt{T}/a)Q(y)$. A standard asymptotic sandwich transfers the equivalent from the model integrand to $Q(e^{-s})$. $\square$

14.3 Large weighted quantile moments

For $p \in \mathbb{N}_0$, define

$$q_p = \int_0^1 u^p Q(u) du, \quad C_p = \int_0^1 (1-u)^p Q(u) du. \quad (117)$$

Reflection (88) gives the exact identity

$$q_p = \frac{1}{p+1} - C_p. \quad (118)$$

Theorem 14.3 (Large quantile-moment correction). *As $p \rightarrow \infty$ through the nonnegative integers,*

$$C_p \sim \frac{Q(1/p)}{p}, \quad (119)$$

and therefore

$$q_p = \frac{1}{p+1} - \frac{\sqrt{\log p}}{e\sqrt{\log 2}p} \exp\left[-\sqrt{2\log 2 \log p}\right] (1 + o(1)). \quad (120)$$

Proof. Set $u = t/p$. Then

$$\frac{pC_p}{Q(1/p)} = \int_0^p \left(1 - \frac{t}{p}\right)^p \frac{Q(t/p)}{Q(1/p)} dt.$$

For fixed $t > 0$, the integrand tends to e^{-t} by slow variation. Potter bounds and the elementary exponential bound on the binomial kernel supply domination after splitting the integral into a compact interval and a tail. The limit is $\int_0^\infty e^{-t} dt = 1$. Substitute (109) at $y = 1/p$ and use (118). $\square$

Let $M_n = \max(X_1, \dots, X_n)$ and $m_n = \min(X_1, \dots, X_n)$. From (94) and symmetry,

$$1 - \mathbb{E}M_n = \mathbb{E}m_n = \int_0^1 F(t)^n dt. \quad (121)$$

The theorem immediately gives a compact order-statistic equivalent.

Corollary 14.4 (Extreme order statistics and high inverse order). *As $n \rightarrow \infty$ through the positive integers,*

$$\mathbb{E}m_n = 1 - \mathbb{E}M_n \sim Q(1/n). \quad (122)$$

Consequently,

$$(\mathcal{J}_n Q)(1) = \frac{1}{n!} \int_0^1 F(t)^n dt \sim \frac{Q(1/n)}{n!}. \quad (123)$$

The correction $Q(1/n)$ is stretched-logarithmic: it is smaller than every negative power of $\log n$, but larger than $n^{-\varepsilon}$ for every fixed $\varepsilon > 0$. This scale is invisible to a conventional expansion in powers of $1/n$.

15 Numerical experiments and reproducibility

The archive accompanying this report contains `numerical_experiments.py`, a fully commented Python script using only the standard library and `mpmath`. It computes two independent moment systems:

$$(n+1)(2^n-1)d_n = \sum_{k < n} \binom{n+1}{k} d_k, \quad (124)$$

$$(4^n-1)c_n = \sum_{k < n} \binom{2n}{2k} \frac{c_k}{2(n-k)+1}. \quad (125)$$

The second recurrence follows from $H(2z) = (\sinh z/z)H(z)$ for the centered moment generating function and is independent of the first recurrence. Exact rational arithmetic verifies (45) for $0 \leq p \leq 8$ and (47) for $1 \leq n \leq 6, 0 \leq p \leq 8$.

Table 2: Independent evaluations of $\mathcal{A}(\alpha) = \int_0^1 x^\alpha F(x) dx$. The first column uses the Newton series (42); the second uses (41) and the centered expansion (52). The shown discrepancies are truncation errors, not statistical errors.

α	Newton–Fabius series	Centered-moment formula	absolute difference
0.5	0.4209681733402671557991	0.4209681733402671557990	9.0×10^{-23}
−0.5	0.6073863107431723970836	0.6073863107431723973782	2.9×10^{-19}
−1	0.7582400004044086356798	0.7582400004044086568776	2.1×10^{-17}
−2	1.3148861440057837837803	1.3148861440058189655548	3.5×10^{-14}
2.3	0.2591812728638398877191737152	0.2591812728638398877191737148	3.7×10^{-28}

The script also checks the exponential master identity at $z = 0.7 + 0.2i$. The moment-series and independent infinite-product values of $G(z)$ agree to approximately 24 decimal places, and the two sides of (57) agree to the same precision. Full machine-readable values are in `verification_output.txt` and `fractional_power_checks.csv`.

Numerical interpretation

The positive series for negative powers converge more slowly as the negative exponent grows. The displayed digits for $\alpha = -2$ are therefore fewer than for positive or fractional exponents. The code reports both truncations and their discrepancy rather than presenting unsupported extra digits.

16 Conjectures and frontier directions

16.1 A periodic saddle expansion for weighted primitives

The exact Newton–Volterra series and the repository’s all-orders periodic Lambert expansion should combine into an all-orders local expansion of (34).

Conjecture 16.1 (Weighted primitive saddle hierarchy). *For fixed $n \geq 1$ and $\alpha \in \mathbb{C}$, let $\lambda_n(x)$ be the exact Lambert phase associated with $x/2^n$. There exist smooth one-periodic functions $P_{j,n,\alpha}$ such that, for every $N \geq 1$,*

$$\frac{\mathcal{J}_n[t^\alpha F(t)](x)}{x^\alpha 2^{\binom{n}{2}} F(x/2^n)} = 1 + \sum_{j=1}^{N-1} \frac{P_{j,n,\alpha}(\lambda_n(x))}{\lambda_n(x)^j} + O(\lambda_n(x)^{-N}). \quad (126)$$

The first coefficient is obtained by inserting the phase expansion of $\mathcal{J}_{n+1}F/(x\mathcal{J}_nF)$ into (35).

This conjecture is structurally plausible because each successive term of (23) shifts the exact phase by approximately one unit, and the periodic fluctuation is therefore sampled coherently rather than averaged away.

16.2 Second-order large-quantile correction

Ignoring the next periodic inverse correction for a moment, expansion of the explicit leading model in theorem 14.3 gives

$$\frac{Q(t/p)}{Q(1/p)} = 1 + \frac{a \log t}{2\sqrt{\log p}} + O\left(\frac{1 + |\log t|^2}{\log p}\right).$$

Since $\int_0^\infty e^{-t} \log t \, dt = -\gamma$, this suggests the following conditional refinement.

Conjecture 16.2 (Universal first correction, modulo periodic inverse jets). *If the relative error in (109) is uniformly $o((\log(1/y))^{-1/2})$ on fixed multiplicative windows, then*

$$\frac{pC_p}{Q(1/p)} = 1 - \frac{\gamma\sqrt{2\log 2}}{2\sqrt{\log p}} + o((\log p)^{-1/2}). \quad (127)$$

In the full problem, the repository's periodic saddle jets may add a bounded oscillatory term at this same or the next scale; determining the precise interaction is open.

16.3 Large negative moments

The positive moment function $M(s)$ is entire, so all negative moments $N(r) = M(-r) = \mathbb{E}X^{-r}$ exist. Exact integration by parts gives

$$N(r) = 1 + r \int_0^1 x^{-r-1} F(x) \, dx. \quad (128)$$

Inserting the repository's exact-phase expansion and applying Laplace's method in the Lambert variable leads to the next conjecture.

Conjecture 16.3 (Periodic large-negative-moment law). *As real $r \rightarrow +\infty$,*

$$\log M(-r) = \frac{\log 2}{2} r^2 - r \log r + \left(1 + \frac{\log 2}{2}\right) r + \frac{1}{2} \log r + C + \Omega(r) + o(1), \quad (129)$$

where C is constant and Ω is a bounded smooth one-periodic function determined by the saddle fluctuation Ψ . For integral r , the periodic phase is sampled at a fixed saddle class and may collapse into the constant term.

The quadratic coefficient and the terms through $\frac{1}{2} \log r$ follow from the stationary phase near $\lambda = r + 1/2$; the exact periodic saddle shift must still be controlled.

16.4 Holonomicity, arithmetic, and formalization problems

The following directions appear especially promising.

1. **Lean formalization.** The reusable first wave is complete: `FabiusFunction.Normalize` `dVolterra` supplies the Banach-valued operator, the literal-iteration bridge, and the finite polynomial/monomial commutators; `FabiusFunction.FabiusAntiderivatives` supplies the signed and bounded integer ladders and their natural-monomial consequences. The next finite targets are the exterior $x > 1$ branch and matching jets in (37), followed independently by the inverse identity (89). The normal-convergence proof of (23), its Mellin consequences, and fractional order form a separate analytic wave.

2. **Dyadic arithmetic.** Substitute the repository's exact q -binomial evaluator into (24). Determine sharp 2-adic valuations and odd denominator factors of $H_{n,p}(a/2^m)$. The two-parameter identity (47) may create cancellations not visible term by term.
3. **Newton-series uniqueness.** Establish optimal growth classes in which (42) is the unique Newton interpolation of $(1 - d_{p+1})/(p + 1)$. The superpolynomial decay of d_n should yield stronger convergence than generic Carlson-type hypotheses.
4. **Non-holonomicity.** Test the conjecture that $\mathcal{A}(\alpha)$ and $M(\alpha)$ are not D-finite and that their integer samples are not P-recursive. The dyadic functional equation for G , the infinite Bernoulli cumulant sequence, and the periodic saddle fluctuation all point in this direction, but none alone is a proof.
5. **Fractional order and Mellin–Barnes methods.** Develop (84) as a two-complex-variable special function in (α, ν) . Its endpoint form (86) may admit Mellin–Barnes continuation and residue expansions tied to the sinc product.
6. **Inverse powers of the CDF.** Study $\int_0^x F(t)^s dt$, the sole new object in (91). For positive integral s it represents order statistics; near $s = 0$ its one-sided derivatives involve $\int (\log F)^k$, whose growth should encode the quadratic-log endpoint.
7. **Fourier antiderivatives.** The complete transform (60) is an infinite sinc product, while the exterior branches of (73) are moment polynomials. A distributional Fourier theory that explicitly separates those polynomial tails from $(2\pi i\xi)^{-n} \widehat{\text{up}}(\xi)$ should produce useful compact renormalized primitives.
8. **Order statistics and periodic corrections.** Push theorem 14.3 to all orders using the closed periodic inverse jets already present in the frontier volume. This would transfer the Lambert–W and periodic structure into a new probabilistic observable: $\mathbb{E} \min(X_1, \dots, X_n)$.

17 Summary of principal formulas

Object	Exact formula, leading result, and current Lean status
Generic polynomial weight	Finite Hasse-derivative formula (18); Lean formalized for normed-space-valued inputs, arbitrary base and endpoint, and every $n \in \mathbb{N}_0$.
n -fold primitive of F for $x \leq 1$	$\mathcal{J}_n F(x) = 2^{\binom{n}{2}} F(x/2^n)$, Lean formalized for every $n \in \mathbb{N}_0$. The signed analogue is formalized for every real x .
Natural monomial weight	Finite sum (25) is Lean formalized for $x \leq 1$; the $x > 1$ global completion in (37) remains a frontier.
Complex monomial weight	Absolutely convergent Newton–Volterra series (23), entire in α ; not yet Lean formalized.
Complete endpoint	$\int_0^1 x^\alpha F(x) dx = (1 - M(\alpha + 1))/(\alpha + 1)$ and Newton series (42); Mellin/Newton frontier.
Negative integer weight	Positive dyadic series (33); logarithmic constant (51)–(53); frontier.
Reflection/complement	$F(1-x) = 1 - F(x)$, and $\mathcal{J}_n[x^p(1-F)] = p!x^{p+n}/(p+n)! - H_{n,p}$.
Rvachev up-function	Complete four-branch formula (73); exterior branches are moment polynomials; folded-primitive frontier.
Higher derivatives	Thue–Morse sums (75) and (76); single dilation for $\mathcal{F}$.
Fractional order	Probability formula (81) and commutator (84); frontier.

Object	Exact formula, leading result, and current Lean status
Inverse first primitive	$\int_0^y Q(u) du = yQ(y) - F(Q(y)/2)$, as in (89); frontier.
Inverse arbitrary order	$\mathcal{I}^\nu Q(y) = \Gamma(\nu + 1)^{-1} \int_0^{Q(y)} (y - F(t))^\nu dt$; frontier.
Inverse endpoint asymptotic	Weighted Karamata law (112), logarithmic boundary (116), and large moment correction (120); frontier.

A Corpus comparison map

Source family	Role in the nonduplication audit
Fabius_Function_and_Rvachev_Up/	Primary live exposition: bounded/signed conventions, shape, derivatives, moments, dyadic arithmetic, effective flatness, sharp Lambert asymptotics, and formalization status.
FabiusFunction_Mathematical_Glossary/	Notation and cross-reference normalization; used to prevent collisions among F , up, the signed extension, moments, and inverse conventions.
fabius_lean_walkthrough/	Map from human statements to Lean declarations; used to identify which imported facts are already formalized.
Non_Elementarity_of_the_Fabius_Function/	Analytic-locus and non-elementarity results; establishes why the native finite Fabius formula, rather than an elementary primitive, is the appropriate notion of closed form.
Consolidated frontier volume	Consolidated frontier volume: repeated averaging, inverse function, small-argument inversion, q-calculus, sinc-product decay, Lambert–W methods, periodic saddle jets, and other deferred claims.
Frontier draft packages	Package-level checks of inverse, exponent-sequence/Newton, repeated-integration, Fourier-decay, Thue–Morse, and asymptotic reports. The Newton report concerns exponent sequences and q-Richardson reconstruction, not Newton series of monomial antiderivatives.
papers/	Source transcriptions and corrected papers, especially Arias de Reyna’s compact support and arithmetic articles; used for historical formulas and moment normalizations.
archive/	Frozen early notes, primary-exposition revisions, q-calculus studies, and standalone reports. Historical descendants were treated as lineages rather than independent novelty claims.

The decisive negative checks were not based only on titles. The live primary source and consolidated frontier source were searched for the literal phrases “antiderivative” and “indefinite integral”; neither contains a treatment of the present problem. The repeated-integration material concerns iterative averaging/convergence to up, not the Volterra primitives here. The inverse material treats regularity, dyadic inversion, non-elementarity, and endpoint asymptotics, but not (89), (97), or weighted quantile integrals. The Newton draft treats exponent sequences, sinc products, Prouhet cancellation, and q-Richardson reconstruction, not (42). These are snapshot-relative source-document comparisons. The later Lean modules listed in the status crosswalk now formalize the integer Volterra core, while the complex, fractional, Mellin, folded, inverse, and exterior-piecewise claims remain outside that formalized core.

B Exact initial values and sample antiderivatives

The first moments computed from (7) are

$$\begin{aligned} d_0 &= 1, & d_1 &= \frac{1}{2}, & d_2 &= \frac{5}{18}, & d_3 &= \frac{1}{6}, \\ d_4 &= \frac{143}{1350}, & d_5 &= \frac{19}{270}, & d_6 &= \frac{1153}{23814}, & d_7 &= \frac{583}{17010}. \end{aligned}$$

The first centered moments are

$$c_0 = 1, \quad c_1 = \frac{1}{9}, \quad c_2 = \frac{19}{675}, \quad c_3 = \frac{583}{59535}, \quad c_4 = \frac{132809}{32531625}.$$

For every real $x \leq 1$, the first ordinary primitives are

$$\int_0^x F(t) dt = F(x/2), \quad (130)$$

$$\int_0^x tF(t) dt = xF(x/2) - 2F(x/4), \quad (131)$$

$$\int_0^x t^2 F(t) dt = x^2 F(x/2) - 4xF(x/4) + 16F(x/8), \quad (132)$$

$$\int_0^x t^3 F(t) dt = x^3 F(x/2) - 6x^2 F(x/4) + 48xF(x/8) - 384F(x/16). \quad (133)$$

At $x = 1$, these equal respectively

$$\frac{1}{2}, \quad \frac{13}{36}, \quad \frac{5}{18}, \quad \frac{1207}{5400}. \quad (134)$$

For a first negative-power example,

$$\int_0^x \frac{F(t)}{t} dt = \sum_{r \geq 0} r! 2^{\binom{r+1}{2}} x^{-r-1} F\left(\frac{x}{2^{r+1}}\right). \quad (135)$$

For a fractional example,

$$\int_0^x t^{1/2} F(t) dt = \sum_{r \geq 0} (-1)^r 1/2^r 2^{\binom{r+1}{2}} x^{1/2-r} F\left(\frac{x}{2^{r+1}}\right). \quad (136)$$

C Derivation of the centered-moment recurrence

Let

$$H(z) = \mathbb{E}e^{z(2X-1)} = \sum_{n \geq 0} c_n \frac{z^{2n}}{(2n)!}. \quad (137)$$

The distributional fixed point

$$2X - 1 \stackrel{d}{=} \frac{V + (2X' - 1)}{2}, \quad (138)$$

where V is uniform on $[-1, 1]$ and X' is an independent copy of X , gives

$$H(2z) = \frac{\sinh z}{z} H(z). \quad (139)$$

Equating coefficients of z^{2n} yields

$$(4^n - 1)c_n = \sum_{k=0}^{n-1} \binom{2n}{2k} \frac{c_k}{2(n-k)+1}, \quad (140)$$

which is the independent recurrence used in the numerical checks.

D Implementation notes

The supplied Python program has four layers.

1. Exact Fraction arithmetic generates the initial d_n and c_n , checks the terminating natural-power identity, and verifies the two-parameter identity with no rounding.
2. High-precision mpmath recurrences generate longer moment arrays.
3. The Newton series (42) and centered formula (52) are evaluated independently for positive, fractional, negative, and complex exponents.
4. The moment series for G is compared with the independent product (10), after which (57) is checked.

No Monte Carlo sampling is used. The default parameters in the packaged output use 85 decimal digits, $d_0, \dots, d_{1500}$, and $c_0, \dots, c_{700}$.

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